

Part II: Assignment #4

Transmissions from the US to the European Stock Market

2.1 Introduction

This paper studies fundamental characteristics of the European stock index, Stoxx 600. In addition we investigate the potential spill over effects from the American traded S&P500 index on Stoxx 600. To conduct this analysis we utilize the GARCH framework to model the conditional variance and mean of the within trading day return.

Later on we extend the model to a threshold GARCHX model and include lagged values of the within trading day return of the SP500 index. Specifically we estimate an ADL(2,1)-GARCH(1,1)X model using quasi ML estimation. Applying misspecification tests we conclude that the model does not indicate misspecification.

The empirical results point to a positive return spillover from the S&P500 to the Stoxx 600 index, which means that an increase in the within trading day return of the SP500 in the previous period has a positive impact on the within trading day return of the Stoxx 600 in the current period with an elasticity of about 0.2. However, we find a less clear volatility spillover as we estimate a borderline significant parameter with low economic significance.

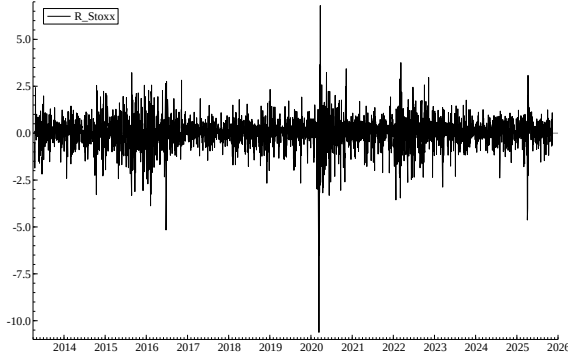
2.2 Presentation of data

In this paper we consider the within trading day returns of the European Stoxx 600 index, covering the period from April 15 2015 to November 13 2025. Furthermore, we use the within trading day returns of the American S&P500 index to determine whether or not this index is helpful in predicting the conditional variance and mean of the Stoxx 600 index. We define the within trading day returns as follows

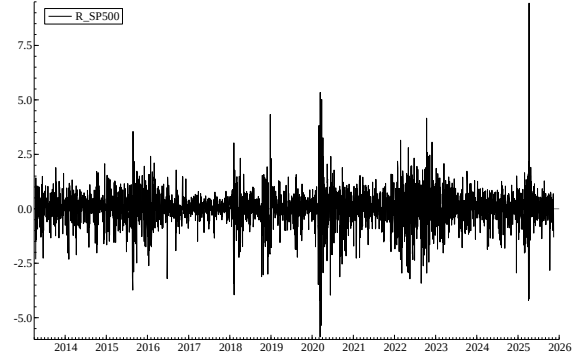
$$R_Stoxx_t = 100 \cdot [\text{Log}(\text{Close_Stoxx}_t) - \text{Log}(\text{Open_Stoxx}_t)] \quad (2.1)$$

$$R_SP500_t = 100 \cdot [\text{Log}(\text{Close_SP500}_t) - \text{Log}(\text{Open_SP500}_t)] \quad (2.2)$$

From figure 4 it is clear that the (within trading day) returns of the two stock indexes exhibit well known stylized characteristics of financial time series. There is clear volatility clustering and fluctuations orders of magnitude larger than the standard fluctuations. This was for instance the case during the Covid 19 pandemic, but also later in the context of the Russian invasion of Ukraine and the following the rise in energy prices. Therefore the natural modelling framework will be the ARCH-family of time series models. In the econometric analysis that follow we also find additional standard features of stock returns: asymmetric effects of returns on future volatility, and no statistically or economically significant predictability in the conditional mean returns.



(a) Stoxx 600



(b) SP500

Figure 4: Characterization of the within trading day returns

2.3 Econometric theory

2.3.1 GARCH and TGARCH

Section 2.2 argues that the data exhibits substantial volatility clustering. GARCH models try to take this into account by modelling the time-varying variance as an autoregressive (or ADL) process. Specifically, we model the conditional time varying variance and the conditional mean. We consider the threshold GARCH, TGARCH(p,q), process given by

$$y_t = x_t' \beta + \epsilon_t \quad (2.3)$$

$$\epsilon_t = \sigma_t z_t, \quad z_t | \mathcal{I}_{t-1} \stackrel{i.i.d}{\sim} (0, 1) \quad (2.4)$$

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^p \tau_i \epsilon_{t-i}^2 I(\epsilon_{t-i} < 0) + \sum_{j=1}^q \beta_j \sigma_{t-j}^2, \quad \alpha_i, \beta_i, \tau_i \geq 0 \forall i, j, \omega > 0 \quad (2.5)$$

, where $\mathcal{I}_{t-1} = \{x_1, y_{t-1}, x_{t-1}, y_{t-2}, x_{t-2}, \dots\}$ defines the information set. The conditional mean $x_t' \beta$ potentially contains all or some of the following: lagged covariates, other explanatory variables and dummies. The residuals ϵ_t are defined by the product between the conditional standard deviation σ_t and the independently and identically distributed innovations z_t . The innovations take on an arbitrary distribution with mean zero and variance 1. From (2.5) we can interpret the modelling of the conditional variance. Shocks to the process will propagate through the conditional variance determined by the coefficients. Including exogenous variables in eq. (2.5) the unconditional variance for a symmetric distributed z_t is given by:

$$\sigma^2 = \frac{\omega + \delta}{1 - \sum_{i=1}^p \alpha_i - \frac{1}{2} \sum_{i=1}^p \tau_i - \sum_{j=1}^q \beta_j} \quad (2.6)$$

where δ contains the coefficients on the exogenous variables. From (2.6) we note that the unconditional variance exists for

$$\sum_{i=1}^p \alpha_i + \frac{1}{2} \sum_{i=1}^p \tau_i + \sum_{j=1}^q \beta_j < 1 \quad (2.7)$$

, which is a sufficient stationarity condition for the process. To investigate asymmetric effects in the GARCH framework, we extend the model. Namely we utilize the TGARCH and TGARCH X models, which allow for asymmetric volatility responses to innovations. Specifically this alters (2.5) and we have the following general ADL(A,B)-TGARCH(p,q)-X(w) model specification given by,

$$R_Stoxx_t = \sum_{a=1}^A \phi_a R_Stoxx_{t-a} + \sum_{b=1}^B \mu_b R_SP500_{t-b} \quad (2.8)$$

$$\epsilon_t = \sigma_t z_t, \quad z_t | \mathcal{I}_{t-1} \stackrel{i.i.d}{\sim} (0, 1) \quad (2.9)$$

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^p \tau_i \epsilon_{t-i}^2 I(\epsilon_{t-i} < 0) + \sum_{j=1}^q \beta_j \sigma_{t-j}^2 + \sum_{k=1}^w \zeta_k R_SP500_{t-k} \quad (2.10)$$

Note that for most practical purposes, it is sufficient to estimate a GARCH(1,1) model since σ_t^2 exhibit a long lasting memory. The conditional variance is required to be non-negative and therefore we impose the following restrictions on the estimated parameters. In line with (2.5) we require $\alpha_i, \beta_j, \zeta_k, \tau_i \geq 0 \forall i, j, k$ and $\omega > 0$. Note that these restrictions imply difficulties when trying to test the parameters in the conditional variance equation for statistical significance, either because 0 lies outside the parameter space or because the test statistics follow non-standard distributions.

2.3.2 (Quasi) Maximum Likelihood Estimation

To estimate the parameters of the model will need to specify a distributional shape of ϵ_t . Cf our earlier discussion we want to allow for a larger proportion of extreme observations (compared to a normal distribution), and we therefore choose the students t-distribution. Formally we let $z_t | \mathcal{I}_{t-1} \stackrel{i.i.d}{\sim} t(0, 1, \nu)$ or equivalently $\epsilon_t | \mathcal{I}_{t-1} \stackrel{i.i.d}{\sim} t(0, \sigma_t^2, \nu)$, where ν is the degrees of freedom. By scaling the standard t-distribution with $\sqrt{(\nu-2)/\nu}$ we normalize so that $Var(z_t) = 1$ for $\nu > 2$. We obtain the following scaled students $t(\nu)$ log likelihood function.

$$\log L_T(\theta) = \sum_{t=1}^T \log \ell_t(\theta) = \sum_{t=1}^T \log \left[\frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\pi(\nu-2)\sigma_t^2(\theta)}} \left(1 + \frac{(y_t - \mu_t(\theta))^2}{(\nu-2)\sigma_t^2(\theta)} \right)^{-\frac{\nu+1}{2}} \right] \quad (2.11)$$

The (quasi) maximum likelihood estimator is then defined as

$$\hat{\theta}_{QMLE} = \arg \max_{\theta \in \Theta} \log L_T(\theta) \quad (2.12)$$

Under certain high level assumptions⁹ and given that the score $s_t(\theta_0) = \frac{\partial}{\partial \theta} \ell_t(\theta)|_{\theta=\theta_0}$ satisfies $\mathbb{E}[s_t(\theta_0)] = 0$ the estimator is consistent such that $\hat{\theta}_{QMLE} \xrightarrow{p} \theta_0$, where $\theta_0 \in \Theta$ is the population parameters identifying the true process, and asymptotically normal such that

$$\sqrt{T}(\hat{\theta}_{QMLE} - \theta_0) \xrightarrow{d} N(0, \Omega_{\theta}^*(\theta_0)), \quad T \rightarrow \infty \quad (2.13)$$

where the asymptotic variance is a function of the true parameter $\theta_0 \in \Theta$, and can be shown to have the formula

$$\Omega_{\theta}^*(\theta_0) = \mathcal{I}(\theta_0)^{-1} \mathcal{J}(\theta_0) \mathcal{I}(\theta_0)^{-1} \quad (2.14^{10})$$

In the case of the model distribution correctly specifying the distribution of the data-generating process $\mathcal{I} = \mathcal{J}$ and $\Omega_{\theta}^*(\theta_0) = \mathcal{I}(\theta_0)^{-1}$ and the presented estimator is efficient.

2.3.3 Misspecification

We test for misspecifications of the model, specifically testing for no-ARCH effects and autocorrelation. The no-ARCH test is carried out by regressing the squared residuals on lagged values of the squared residuals¹¹.

$$\hat{\varepsilon}_t^2 = \gamma_0 + \gamma_1 \hat{\varepsilon}_{t-1}^2 + \gamma_2 \hat{\varepsilon}_{t-2}^2 + \cdots + \gamma_p \hat{\varepsilon}_{t-p}^2 + \text{error} \quad (2.15)$$

The null hypothesis of no ARCH effects is

$$H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_p = 0 \quad (2.16)$$

and is tested using an LM test statistic, which under the null follows a $\chi^2(p)$ distribution. We also employ a normality test to test the the Gaussian MLE. See table 2 for the null hypothesis and the distribution of the tests for no-autocorrelation.

⁹See Nielsen (2025), Assumption 3.1

¹⁰Where $\mathcal{I}$ is the fisher-information $\mathcal{I}(\theta_0) = -\mathbb{E} \left[\frac{\partial^2 \log \ell_t(\theta_0)}{\partial \theta \partial \theta'} \right]$ and $\mathcal{J}(\theta_0) = \mathbb{E} \left[\frac{\partial \log \ell_t(\theta_0)}{\partial \theta} \frac{\partial \log \ell_t(\theta_0)}{\partial \theta'} \right]$

¹¹Nielsen (2025), section 12.3

¹¹See section 2.3 "Formulation and Misspecification Testing," Nielsen (2025) for an explanation of the degrees of freedom for the test statistics asymptotic distributions.

Test	Null hypothesis	Asymptotic distribution (under $\mathcal{H}_0$)
No-ARCH test ^a	no ARCH effects	$\chi^2(q)$
Portmanteau test ^b	no autocorrelation	$\chi^2(h - p)$
Normality test ^c	normally distributed residuals	$\chi^2(2)$

^a Nielsen (2025), p. 273.

^b Slides 5: Model Formulation and Misspecification Testing.

^c Nielsen (2025), p. 44.

Table 2: Overview of misspecification tests and their null hypotheses

2.4 Empirical results

2.4.1 Model selection and misspecification tests

In order to model the Stoxx 600 index and characterise the conditional mean and variance, we estimate a baseline AR(0) – TGARCH(1, 1)-model in accordance with (2.9)-(2.10).¹² Note for any initial general autoregressive order, a joint robust Wald test implies that we cannot reject the null hypothesis, that all lagged variables are insignificantly different from zero, at any conventional significance level. This implies that the past returns have no linear predictability regarding the current returns, as expected.

The estimated model appears well-specified. As table 3 (columns 2-5) shows, we cannot reject the null of no ARCH effects and autocorrelation.

Test	TGARCH (t)	TGARCH X (t)	TGARCH X (t)*	TGARCH X (t)**
No ARCH test	0.39284 (0.6752)	0.55601 (0.5735)	1.0579 (0.3474)	0.77001 (0.4632)
Portmanteau test	68.828 (0.1355)	65.233 (0.2123)	54.143 (0.0820)	49.361 (0.1236)

p values in parentheses. All models assume student-t distributed innovations.

* Sample 2013-04-16 to 2019-12-31. ** Sample 2019-12-31 to 2025-11-13.

Table 3: Misspecification tests for the TGARCH models

To examine whether within trading day returns of the SP500 index help predict within trading day returns of the Stoxx 600 index, we extend model to an ALD(0,1)-TGARCH(1,1) X model, as described in Section 2.3.1. Table 3 (column 3) shows the misspecification test regarding the TGARCH X model and the model does not indicate misspecification.

¹²We also estimate the parameters based on z_t being conditionally standard normal distributed. The null of normality is strongly reject, and hence we carry out the main analysis as described in section 2.3 using the t-distribution. Note that we only require $\mathbb{E}[\frac{\partial}{\partial \theta} L_t(\theta) | \theta = \theta_0] = 0$ along with the high level assumptions mentioned in section 2.3 for asymptotically normally distributed estimates. However the distributional choice still affects efficiency of the estimator.

2.4.2 Estimation results - Baseline TGARCH

Table 4 (column 1) reports the estimates from the baseline TGARCH model for the within trading day returns of the STOXX 600 index. The constant for the conditional mean is estimated at 0.04975, implying an average within trading day return of approximately 5%. The conditional variance constant, $\omega = 0.03136$, is positive and statistically significant but small relative to the strong persistence component of the models conditional variance.: $\beta = 0.81713$. The coefficient α is restricted to be zero (or above) with reported value $\alpha = 0.00000$, suggesting that positive innovations have little impact on the next periods volatility - beyond what is captured by the lagged conditional variance. In contrast, the threshold parameter $\tau = 0.25648$ is positive and statistically significant, indicating that negative innovations lead to a substantial increase in conditional variance. Note as well that $\alpha + \beta + \frac{1}{2}\tau = 0.945$, is below 1, implying that volatility is highly persistent yet covariance-stationary. Overall, the results provide strong evidence of asymmetric volatility dynamics and are consistent with the existing literature on stock returns.

Parameter	TGARCH	TGARCH X	TGARCH X*	TGARCH X**
Constant	0.04975 (0.011)	0.03611 (0.010)	0.02904 (0.014)	0.04715 (0.016)
$R_{SP500_{t-1}}$		0.16905 (0.017)	0.16487 (0.028)	0.17268 (0.022)
$R_{SP500_{t-1}}^2$		0.03360 (0.011)	0.03397 (0.019)	0.04099 (0.016)
ω	0.03136 (0.007)	0.02927 (0.006)	0.02844 (0.009)	0.03097 (0.009)
α	0.00000	0.00000	0.00000	0.02432 (0.026)
β	0.81713 (0.024)	0.78703 (0.025)	0.79601 (0.046)	0.75376 (0.038)
τ (Threshold)	0.25648 (0.048)	0.25874 (0.043)	0.25163 (0.066)	0.25364 (0.063)
student t df	5.75327 (0.555)	6.04170 (0.609)	6.42986 (0.973)	5.59353 (0.744)
$\alpha + \frac{1}{2}\tau + \beta$	0.945	0.916	0.922	0.905
Observations	3256	3255	1734	1522

Robust standard errors in parentheses. * 2013-04-16 to 2019-12-31. ** 2019-12-31 to 2025-11-13.

Table 4: Estimation results for TGARCH models

2.4.3 Estimation results - Extended TGARCH X

The extended model (see column 2 table 4) now includes the lagged SP500 within trading day returns in the mean equation and the squared SP500 within trading day returns in the variance equation. The estimate for $R_{SP500_{t-1}}$ shows an economically significant return spillover from SP500 to Stoxx 600 (0.16905). In practice this means that a 1 pp. increase in the within trading day returns of the SP500 in period $t - 1$ entails a positive increase to the within trading day returns of the Stoxx 600 on 0.17 pp. in period t .

The volatility spillover from SP500 to Stoxx 600 is borderline significant and economically modest. A 1 pp. shock to the squared within trading day returns of the SP500 in period $t-1$ results in a 0.036 pp. increase in the conditional variance of R_Stoxx in period t . The remaining parameters are close to the baseline, and are interpreted similarly to the baseline model, with α being totally insignificant. This indicates that the conditional volatility of the within trading day returns of the European stock index, primarily can be described by its own past shocks, and past volatility, potentially with some small level of spillover from the SP500.

To determine whether or not the spillover effects from the American stock market to the European stock market are constant over time, we estimate the TGARCH X model for two sub-periods. The first runs from 2013 to the end of 2019 (pre covid-19), and the second covers the period from the end of 2019 to 2025 (during and post Covid-19). As seen in table 3, the models remain well specified. The results (see table 4) show that the return spillover is remarkably stable, with only slight deviations from the full sample estimate. The volatility spillover effect varies more from the first sub-period to the second. This indicates that the volatility spillover effect may have become stronger in the later years, although the level remains low. Overall, the sub-period analysis confirms the results we have found previously.

2.5 Conclusion

The empirical analysis using our final model TGARCH-X shows that the spillover effects from the American stock market to the European stock market finds a substantial and significant return spillover of 0.16905 and a less impactful, and less significant volatility spillover effect 0.03360. In practise this implies that the within trading day returns of the American SP500 index, yesterday, affects the within trading day returns of the European Stoxx 600 index today. In contrast we find that the volatility of the Stoxx 600 index is primarily driven by persistence in the volatility $\beta = 0.79$ and previous negative shocks $\tau = 0.26$ to the series itself, and is not affected substantially by the SP500 index.

These findings are inline with other literature suggesting a non-negligible return spillover from the leading US stock indexes to Europe, but also other parts for the world. The findings regarding the volatility spillover are also inline with other literature finding that past volatility is more impactful to the present volatility than the volatility originating from other markets (Dimpfl, 2011).